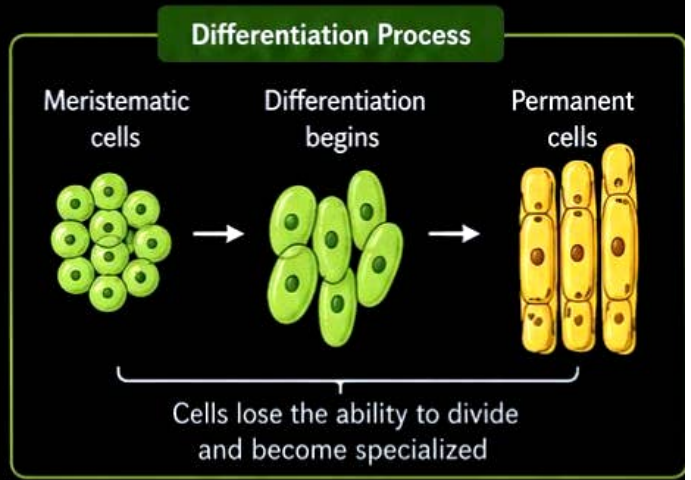


Differentiation

Differentiation = the process by which meristematic tissue becomes specialized (in structure and function) to form permanent tissue — cells lose the ability to divide and take up specific roles like support, transport, or storage.

- Some new cells stay meristematic (keep dividing)
- Other cells differentiate → become **permanent tissue**



3.2.4 Permanent Tissues

Permanent tissues = simple (one cell type) or complex (more than one cell type).

Classified by function into 3 groups:

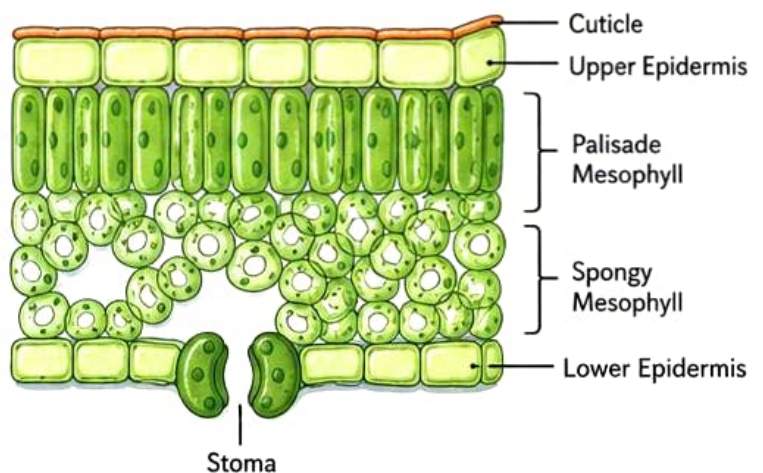
(i) Protective Tissue — Epidermis

- Forms the outermost layer of the plant body
- Single layer of flat, tightly packed, rectangular cells
- Covered by a waxy layer called **cuticle** (secreted by epidermis) — reduces water loss
- In dry-habitat plants, cuticle layer is thicker (less transpiration loss)

Functions of Epidermis:

- Protects against mechanical injury
- Prevents water loss
- Protects against microorganisms/parasites
- Protects against extreme environmental conditions

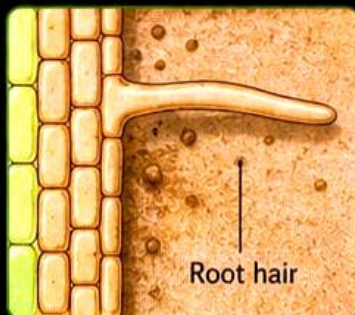
T.S. of Dicot Leaf (showing Epidermis)



Special Modifications

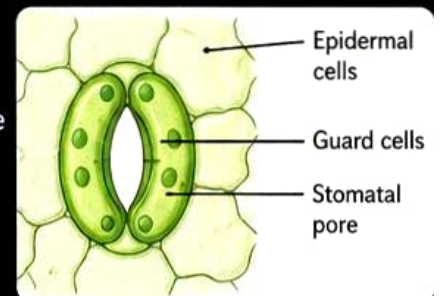
1. Root Hair

Hair-like projections from epidermal cells in roots; increase surface area for absorption of water & minerals.



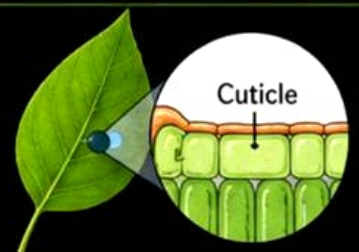
2. Stomata

Pores in leaf epidermis; allow gaseous exchange and transpiration (evaporation of water vapour).

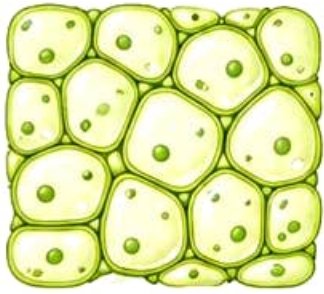


Q: What is the role of the cuticle?

A: The cuticle is a waxy layer over the epidermis that reduces water loss and protects against mechanical injury and invasion by parasites/pathogens.



PARENCHYMA

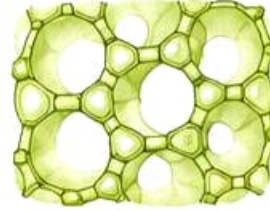


Thin walls

Living

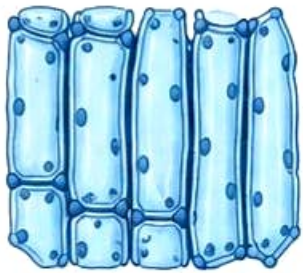


Loosely packed, intercellular spaces present



- Stores food
- Photosynthesis (in green parts)
- Forms air spaces in aquatic plants (helps floating)

COLLENCHYMA

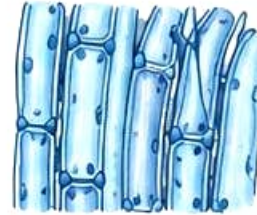


Unevenly thickened (corners thickened due to pectin)

Living

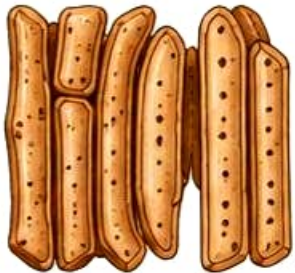


Flexible like rubber



- Gives flexibility & support
- Lets stems/tendrils bend without breaking

SCLERENCHYMA

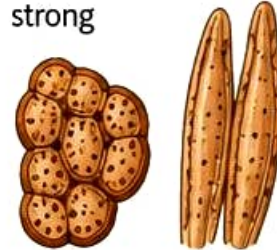


Thick walls (due to **lignin** deposition)

Mostly Dead



Hard and strong



Provides mechanical strength — found in stems, veins, seed coats, nut shells, coconut husk, walnut shell

WHERE DO WE FIND THEM?

1. Parenchyma



Mesophyll of leaf



Storage in fruits



Air spaces in aquatic plants

Key Point:

Loosely packed cells with intercellular spaces.

2. Collenchyma



In leaf stalk (petiole)



In young stems & tendrils

Key Point:

Provides support with flexibility.

3. Sclerenchyma



In stem (wood)



In leaf veins



In nut shells & seed coats

Key Point:

Provides strength and hardness.

Q

Why does a fresh twig bend but a dry twig break?

A

A fresh twig has collenchyma cells with flexible, unevenly thickened walls that provide strength with flexibility, so it bends without breaking. In a dry twig, cells lose water and the tissues become rigid and brittle (mainly due to sclerenchyma), so it breaks.

Fresh Twig



Bends

Dry Twig



Breaks



Remember:

Parenchyma (Store & photosynthesise) • Collenchyma (Support & flexibility) • Sclerenchyma (Strength)



THE ASIAN SCHOOL, DEHRADUN

WORKSHEET

Subject : SCIENCE (BIOLOGY)

Chapter:- Cell: The building block of life

Topic: Cell structure

Class : IX
DDH

MM : 20

Teacher :

	SECTION-A	
1.	The cell membrane is mainly responsible for: A. Producing energy B. Controlling the movement of substances into and out of the cell C. Making proteins D. Storing genetic material	1
2.	The jelly-like substance present between the cell membrane and nucleus is called: A. Cell wall B. Cytoplasm C. Nucleoplasm D. Vacuole	1
3.	Which structure contains the genetic material of a cell? A. Cytoplasm B. Cell membrane C. Nucleus D. Cell wall	1
4.	The cell membrane is described as selectively permeable because it: A. Allows all substances to pass through it B. Does not allow any substance to pass through it C. Allows only certain substances to pass through it D. Allows only water to pass through it	1
5.	When a cell is placed in a solution having a higher concentration of solute than the cell sap, water moves out of the cell. This process is called: A. Diffusion	1

	B. Osmosis C. Plasmolysis D. Endocytosis	
7.	Differentiate between diffusion and osmosis on the basis of the movement of particles and the requirement of a selectively permeable membrane. Give one suitable example of each process.	2
	SECTION-C	
8.	A plant cell is placed in a concentrated salt solution. A. What will happen to the movement of water? B. What will happen to the cell membrane and cytoplasm. C. Name the process responsible for this change.	3
9.	Explain how the nucleus controls the activities of a cell. Also mention the role of chromosomes present inside the nucleus.	3
	SECTION-D	
10	The cell membrane, cytoplasm and nucleus work together to maintain the proper functioning of a cell. Explain this statement by describing: A. The structure and function of the cell membrane. B. The role of cytoplasm in cellular activities. C. The structure and functions of the nucleus. D. How the nucleus and cytoplasm together help in controlling and carrying out cellular activities.	5



THE ASIAN SCHOOL, DEHRADUN WORKSHEET

Subject : SCIENCE (BIOLOGY)

Chapter:- Cell: The building block of life

Topic: The Cell Interior

Class : IX
DDH

Teacher :

MM : 20

SECTION-A

	SECTION-A	
1.	Which cell organelle is known as the “powerhouse of the cell”? A. Ribosome B. Mitochondria C. Golgi apparatus D. Vacuole	1
2.	Which organelle controls most of the activities of the cell? A. Lysosome B. Vacuole C. Cytoplasm D. Nucleus	1
3.	Which organelle is responsible for protein synthesis? A. Mitochondria B. Ribosome C. Golgi apparatus D. Chloroplast	1
4.	Which organelle is mainly involved in packaging and secretion of substances? A. Golgi apparatus B. Nucleus C. Vacuole D. Lysosome	1
5.	A cell contains many chloroplasts and a large central vacuole. Which type of cell is it most likely to be? A. Animal cell	1

	B. Plant cell C. Bacterial cell D. Red blood cell	
	SECTION-B	
6.	What is the function of lysosomes? Why are they sometimes called the “suicidal bags” of the cell?	2
7.	Differentiate between rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) on the basis of their structure and function.	2
	SECTION-C	
8.	A cell is unable to produce enough energy even though It has sufficient food and oxygen. Which cell organelle may not be functioning properly? Identify the organelle and explain how its structure is related to its function.	3
9.	Explain how the endoplasmic reticulum, Golgi apparatus and lysosomes work together to process, transport and manage materials inside a cell.	3
	SECTION-D	
10	A student examines two different types of cells under a microscope. Cell A possesses a cell wall, chloroplasts and a prominent central vacuole, whereas Cell B lacks these structures. Answer the following: A. Identify Cell A and Cell B on the basis of their structural features. B. Name the cell organelle that carries out photosynthesis in Cell A. C. Mention any two functions performed by the large central vacuole in Cell A. D. Describe the role of the nucleus in regulating the activities of both cells. E. Explain how the presence of a cell wall contributes to the structural stability and protection of Cell A.	5



THE ASIAN SCHOOL, DEHRADUN

WORKSHEET

Chapter:- Cell: The building block of life

Topic: Cell Division

Class : IX
DDH

Teacher :

MM : 20

SECTION-A

	SECTION-A	
1.	The process by which a parent cell divides to form new cells is called: A. Respiration B. Cell division C. Nutrition D. Excretion	1
2.	Which biomolecule is the main source of energy for our body? A. Proteins B. Carbohydrates C. Vitamins D. Minerals	1
3.	Proteins are mainly required by the body for: A Growth and repair B. Providing only water C. Maintaining body temperature D. Producing oxygen	1
4.	The cell membrane is described as selectively permeable because it: A. Allows all substances to pass through it B. Does not allow any substance to pass through it C. Allows only certain substances to pass through it D. Allows only water to pass through it	1
5.	During cell division, the genetic material must be properly distributed to the daughter cells so that: A. They can lose their chromosome. B. They receive the required genetic information C. They become non-living D. They stop performing all functions	1
	SECTION-B	
6.	What is cell division? State two reasons why cell division is important for living organisms.	2

7.	Identify any two biomolecules found in cells and mention one function of each.	2
	SECTION-C	
8.	A cell undergoes division and produces two daughter cells. Explain why DNA replication and proper distribution of chromosomes are important before and during cell division.	3
9.	A group of cells in a tissue starts dividing continuously without normal control. A. What condition may develop due to such uncontrolled cell division? B. What is meant by cancer? C. Why can uncontrolled cell division be harmful to the body?	3
	SECTION-D	
10	Cells require both cell division and biomolecules for their growth and functioning. Explain the following: A. Why is cell division necessary for growth and repair B. What is the role of carbohydrates in cells? C. What is the role of proteins in the body? D. What is meant by uncontrolled cell division? E. How can uncontrolled cell division lead to cancer?.	5



THE ASIAN SCHOOL, DEHRADUN WORKSHEET

Chapter:- Tissues in Action
Topic: Plant Tissues

Class : IX
DDH

MM : 20

Teacher :

	SECTION-A	
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1.	Which of the following tissues consists of living cells and generally stores food? A. Sclerenchyma B. Parenchyma C. Xylem D. Epidermis	1
2.	The tissue that provides flexibility to young stems and leaf stalks is: A. Parenchyma B. Collenchyma C. Sclerenchyma D. Internode	1
3.	The epidermis of a plant primarily helps in: A. Transporting food B. Protecting the plant body C. Producing seeds D. Conduction of water	1
4.	A plant part contains cells with unevenly thickened walls, particularly at the corners. Which tissue is most likely present? A. Parenchyma B. Collenchyma C. Sclerenchyma D. Meristematic tissue	1
5.	A student observes a tissue made up of dead cells with thick, lignified walls and very little intercellular space. Identify the tissue and state its major function. A. Parenchyma – storage B. Collenchyma – flexibility C. Sclerenchyma – mechanical strength D. Epidermis – gaseous exchange	1
	SECTION-B	
6.	Give two reasons why a plant tissue is different from animal tissue.	2
7.	What is ground tissue? Mention any one function of ground tissue.	2
	SECTION-C	

8.	Differentiate between Xylem and Phloem on the basis of: A. Any two components B. Functions C. Direction of conduction	3
9.	What is supportive tissue? Explain how supportive tissues are modified to provide mechanical strength and support to plants. Give two examples.	3
	SECTION-D	
10	A plant grows continuously throughout its life because certain regions contain cells that retain the ability to divide repeatedly. These tissues produce new cells, which later differentiate and contribute to the growth and development of various plant parts. Answer the following: A. Name the tissue responsible for the growth of plants. B. Describe the structure of meristematic cells, including any two features that distinguish them from mature permanent cells. C. Name the type of meristematic tissue located at the growing tips of roots and shoots, and state the kind of growth it produces. D. Explain one function of meristematic tissue in the development of a plant.	5



THE ASIAN SCHOOL, DEHRADUN WORKSHEET

Chapter:- Tissues in Action
Topic: Animal Tissues

Class : IX

Teacher : DDH

MM : 20

	SECTION-A	
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1.	Which tissue forms the protective covering of most organs and body surfaces? A. Connective tissue B. Muscular tissue C. Epithelial tissue D. Nervous tissue	1
2.	Which connective tissue connects muscle to bone? A. Ligament B. Tendon C. Cartilage D. Areolar tissue	1
3.	Which type of muscle is involuntary and found in the walls of internal organs such as the intestine? A. Striated muscle B. Smooth muscle C. Cardiac muscle D. Skeletal muscle	1
4.	A tissue is made up of elongated cells that contract and relax to produce movement. Which tissue is being described? A. Nervous tissue B. Epithelial tissue C. Muscular tissue D. Connective tissue	1
5.	A person accidentally touches a hot object and immediately withdraws the hand. Which property of nervous tissue enables this rapid response? A. Secretion B. Contraction C. Transmission of electrical impulses D. Storage of fat	1
	SECTION-B	
6.	Explain fluid connection tissue. Mention one important function of epithelial tissue.	2
7.	Compare between a tendon and a ligament on the basis of their structures and function.	2
	SECTION-C	

8.	Epithelial tissues have different structures to perform different functions. Compare squamous epithelium and columnar epithelium on the basis of their structure, location and function. How does their structure help them perform their functions.	3
9.	Differentiate between squamous epithelium and columnar epithelium on the basis of their structure, location and function.	3
	SECTION-D	
10	The human skeleton is not merely a framework for the body; different parts are specially adapted to provide protection, support and movement. Analyse the role of the following structures: A. The rib cage is made up of ribs and the sternum. Explain how its structure enables it to protect vital organs while allowing the movements necessary for breathing. B. The skull protects the brain, but its bones are not all freely movable. Explain how the structure and arrangement of the skull provide effective protection to the brain. C. Why is the vertebral column made up of several vertebrae instead of being a single rigid bone? D. What is the basic structural and functional unit of nervous tissue?	5

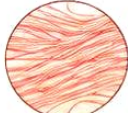


MATRIX – THE DEFINING FEATURE OF CONNECTIVE TISSUE

CLASS 9
SCIENCE

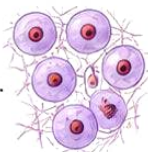
- Matrix is the **non-living intercellular substance** present between the cells of connective tissue.
- In connective tissues, **cells are loosely packed** and embedded in this matrix.

THE MATRIX CAN BE DIFFERENT IN DIFFERENT CONNECTIVE TISSUES

TYPE OF MATRIX	CONNECTIVE TISSUE	FUNCTION	EXAMPLE / LOCATION
 Fluid (Liquid)	 Blood	Transport of materials (gases, nutrients, wastes, hormones) throughout the body.	 Inside blood vessels and heart
 Jelly-like (Flexible)	 Cartilage	Provides flexibility , cushioning and acts as a shock absorber.	 Tip of nose, outer ear, between vertebrae, ends of long bones (joints)
 Hard and Rigid	 Bone	Provides strength , support and protection to the body.	 Skull, ribs, vertebrae, limb bones
 Fibrous	 Tendon & Ligament	Provides great strength and connects one structure to another.	 Tendons connect muscle to bone; ligaments connect bone to bone

WHY IS THE MATRIX IMPORTANT?

- ✓ Holds the cells together.
- ✓ Determines the strength, flexibility and function of the tissue.
- ✓ Gives each connective tissue its special properties.



NCERT KEY CONCEPT

“ Different connective tissues have different types of matrix, and the nature of the matrix decides the function of the tissue. ”



Remember: Same type of cells + **Different type of matrix** = **Different connective tissue** with different function.



FLUID CONNECTIVE TISSUE – BLOOD

CLASS 9
SCIENCE

Blood is the only **fluid connective tissue** in our body.
Its matrix is a fluid called **plasma**.

COMPONENTS OF BLOOD

Component	Description	Function
Plasma (≈55% of blood)	Pale yellow, watery fluid. Contains water, proteins, salts, hormones, enzymes, dissolved nutrients and wastes.	Acts as a transport medium for cells and other substances.
RBCs (Erythrocytes) (≈44% of blood)	Red-coloured cells; contain haemoglobin (iron-rich protein); no nucleus; biconcave disc shape; life span ≈ 120 days.	Carry oxygen from lungs to body cells and some carbon dioxide from cells to lungs.
WBCs (Leukocytes) (<1% of blood)	Larger than RBCs; colourless; have nucleus; different types.	Larger than RBCs; against infections; engulf and destroy harmful microbes.
Platelets (Thrombocytes) (<1% of blood)	Very small cell fragments (not complete cells); no nucleus.	Help in blood clotting at the site of injury to stop bleeding.

EVERYDAY OBSERVATIONS & EXPLANATIONS

Observation	Explanation
A small cut on skin bleeds and then clots.	Platelets release chemicals that convert fibrinogen to fibrin; blood clots and seals the wound.
Infection area becomes red, swollen and sometimes has pus.	WBCs rush to the site to fight microbes; inflammation causes redness and swelling; dead WBCs and microbes form pus.
Face turns red and we breathe faster during exercise.	Muscles need more oxygen; RBCs deliver more oxygen and CO ₂ removal increases; breathing rate rises.
Blood is always red in colour.	Due to haemoglobin in RBCs that gives blood its red colour.

FUNCTIONS OF BLOOD

- Transport of oxygen, carbon dioxide, nutrients, hormones, wastes and heat.
- Defence against infections.
- Helps in blood clotting to prevent excessive blood loss.
- Maintains body temperature and pH balance.



STRUCTURE OF BLOOD



Plasma
(≈55%)

Buffy coat
(WBCs + Platelets)
(<1%)

RBCs
(≈44%)



Plasma – Pale yellow, watery fluid; contains water, proteins, salts, hormones, nutrients, wastes.



WBCs – Defend the body from infections.
Platelets – Help in blood clotting.



RBCs – Contain haemoglobin; transport oxygen.

KEY POINTS

- Blood flows in blood vessels throughout the body.
- Its matrix (plasma) is fluid, so blood can flow easily.
- RBCs, WBCs and platelets are formed in bone marrow.
- Blood is essential for transportation, protection and regulation in the body.



REMEMBER: Fluid matrix (plasma) + Specialised cells (RBCs, WBCs, platelets)
= Fluid connective tissue (**Blood**)

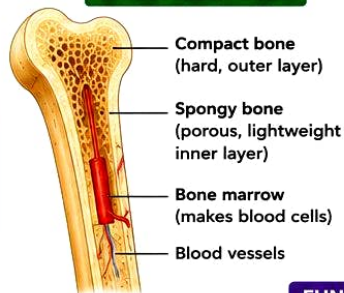
CONNECTIVE TISSUES: BONE AND CARTILAGE

CLASS 9
SCIENCE

A. BONE – A RIGID CONNECTIVE TISSUE

Bone is a hard, rigid connective tissue that forms the skeletal framework of the body.

STRUCTURE OF BONE



FEATURES

- Matrix is hard and rigid.
- Made of calcium and phosphorus salts embedded in a network of collagen fibres.
- Contains bone cells (osteocytes) in small spaces (lacunae).
- Bone marrow in certain bones produces blood cells.

FUNCTIONS



Provides shape, support and protection to the body.



Forms a framework with muscles and helps in movement.



Bone marrow produces blood cells.

LOCATIONS & EXAMPLES



Found in: Skull, ribs, vertebrae, limb bones (e.g., femur, humerus) and other parts of the skeleton.

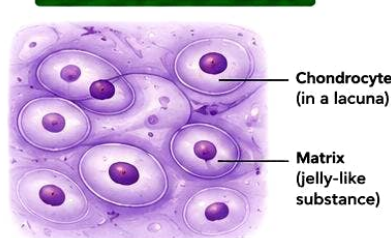
Q. Why are bones hard while blood, from the same tissue category, is a liquid?

A. Bone matrix is solid and mineralised (calcium/phosphorus + collagen) giving hardness and rigidity, whereas blood matrix (plasma) is a fluid.

B. CARTILAGE – A FLEXIBLE CONNECTIVE TISSUE

Cartilage is a flexible connective tissue that provides support and acts as a shock absorber.

STRUCTURE OF CARTILAGE



FEATURES

- Matrix is firm, smooth and jelly-like.
- Made of protein fibres and a gel-like substance.
- Contains cells called chondrocytes in lacunae.
- Not as hard as bone, but flexible.

FUNCTIONS



Provides flexibility and cushioning.



Supports and maintains shape of structures.



Acts as a shock absorber where bones meet.

LOCATIONS & EXAMPLES



Tip of the nose



Outer ear



Between vertebrae (intervertebral discs)



Ends of long bones (joints)



Trachea (windpipe)

Q. Why does pressing your nose or folding your ear not cause pain or breakage, while bending a bone would?

A. Nose and ear are supported by cartilage, which is soft, flexible and returns to its original shape, unlike bone which is rigid.



KEY CONCEPT: Both bone and cartilage are connective tissues, but their matrices are different.
Hard matrix in bone → rigid and strong | Jelly-like matrix in cartilage → flexible and smooth



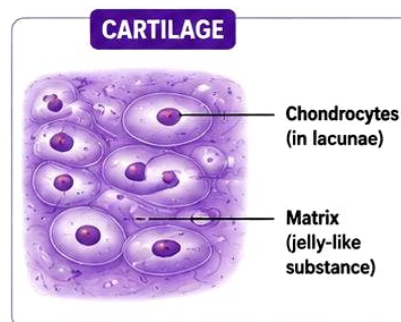
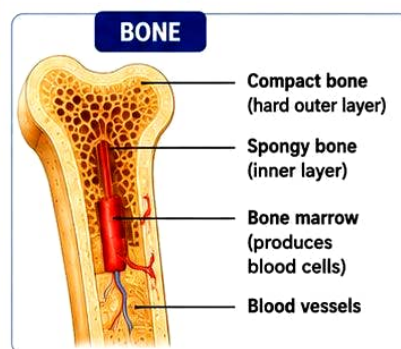
CONNECTIVE TISSUES: BONE vs CARTILAGE

CLASS 9
SCIENCE
(NCERT)

Bone and cartilage are two types of connective tissues.
They differ in their structure and function.



FEATURE	BONE	CARTILAGE
 Nature	Hard and rigid connective tissue	Soft, flexible connective tissue
 Matrix	Hard, mineralised matrix (calcium & phosphorus salts + collagen fibres)	Firm, jelly-like matrix with protein fibres
 Cells	Osteocytes	Chondrocytes
 Function	Provides support, protection and helps in movement	Provides flexibility, cushioning and absorbs shock
 Blood Supply	Rich blood supply	No direct blood supply (nutrients diffuse from surrounding tissues)
 Location	Skull, ribs, vertebrae, limb bones (e.g., femur, humerus) and other parts of skeleton	Tip of nose, outer ear, trachea (windpipe), ends of long bones (joints), between vertebrae (intervertebral discs)
 Flexibility	Very little (almost rigid)	High (flexible and elastic)



NCERT KEY POINTS

- ✓ Bone forms the rigid framework of the body and protects internal organs.
- ✓ Cartilage is softer than bone and prevents friction between bones at joints.



THINK & ANSWER

Why can you bend your ear but not your arm bone?

Answer: The ear contains cartilage, which is flexible, whereas the arm contains bone, which is hard and rigid.



REMEMBER

Bone



= Strength
& Support

Cartilage

= Flexibility
& Cushioning





CONNECTIVE TISSUES: TENDONS AND LIGAMENTS

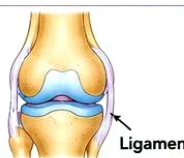
CLASS 9
SCIENCE
(NCERT)

Tendons and ligaments are **dense fibrous connective tissues**.
They provide strong connection and stability in our body.

TENDONS

 DEFINITION	Tendons are tough, fibrous connective tissues that connect muscles to bones .
 STRUCTURE	- Made up of densely packed collagen fibres arranged in parallel bundles. - Very strong and have great tensile strength. - Little ground substance; no blood vessels. 
 FUNCTION	- Transmit the force of muscle contraction to bones. - Help in movement of body parts.
 EXAMPLES / LOCATION	- Tendon of biceps (connects biceps muscle to radius bone of forearm). - Tendon of quadriceps (connects thigh muscle to tibia through patella). 

LIGAMENTS

 DEFINITION	Ligaments are tough, fibrous connective tissues that connect bone to bone at joints.
 STRUCTURE	- Made up of densely packed collagen fibres arranged in parallel bundles. - Very strong but slightly more elastic than tendons. - Little ground substance; no blood vessels. 
 FUNCTION	- Connect bones to bones at joints. - Hold the bones together and provide stability. - Prevent excessive or abnormal movement at joints.
 EXAMPLES / LOCATION	- Ligaments of knee joint (e.g., anterior cruciate ligament). - Ligaments in the ankle, shoulder, elbow joints. 

DIFFERENCES AT A GLANCE

Feature	Tendons	Ligaments
 Connects	Muscle to bone	Bone to bone
 Function	Transmit force of muscle contraction and bring about movement	Hold bones together and provide stability to joints
 Elasticity	Less elastic	More elastic than tendons
 Location	Outside the joint	Inside the joint



KEY POINTS

- ✓ Both are made of dense collagen fibres.
- ✓ They are strong tissues and resist stretching.
- ✓ They work together to allow smooth and safe movements.



 **REMEMBER:** TENDON connects **muscle to bone** and moves the body. | LIGAMENT connects **bone to bone** and stabilizes the joint.